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Robust 3D pose estimation via occlusion-aware training and test time adaptation

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ABSTRACT

Conventional deep learning-based estimators often exhibit limited robustness when exposed to occluded inputs, as their training distributions are biased toward fully visible or minimally occluded human poses. To address this generalization gap, a novel framework, ORPESO (Occlusion Robust Pose Estimation via Synthetic Occlusion), is introduced, explicitly incorporating occlusion diversity into both the learning and inference stages. The proposed method constructs a comprehensive synthetic occlusion space during training by augmenting clean 3D pose sequences with varied and structured occlusion patterns, encouraging the model to learn multimodal spatiotemporal pose representations resilient to partial visibility. Additionally, ORPESO leverages a test-time adaptive recalibration mechanism that performs prediction on rotated samples and averages the prediction of the original and rotated samples. During both training and testing, rotation is applied to facilitate accurate pose recovery under severe occlusions. ORPESO achieves consistent numerical improvements over existing transformer-based methods, achieving average improvements of 15.01 mm (20.6%) in MPJPE on the Human3.6M dataset. On MPI-INF-3DHP, it delivers further gains with PCK improved by 24.2%, AUC by 40.2%, and MPJPE reduced by 57.3% and 27.6% compared to existing state-of-the-art methods, respectively. Extensive evaluations on standard 3DHPE benchmarks, including Human3.6M and MPI-INF-3DHP, demonstrate that ORPESO demonstrates consistent numerical improvements over state-of-the-art baselines.

1. Introduction

With a wide range of applications in action recognition [1], surveillance [2], animation, and human–computer interaction [3,4], and more, 3D Human Pose Estimation (3DHPE) is a fundamental yet difficult task in computer vision. 3DHPE aims to estimate the spatial 3D coordinates of human joints from a single picture or video series [5,6]. Occlusion remains a major obstacle to reliable and accurate pose estimation, even with the remarkable advancements made in deep learning-based techniques in recent years [7]. Estimating the 3D pose of a human from a single camera is a fundamentally ill-posed problem: many different 3D configurations can project to similar 2D appearances. This ambiguity is greatly exacerbated when body parts are occluded. Visual cues that are essential for joint localization are lost or unclear when there is occlusion, whether as a result of self-body overlap, environmental clutter, or object interference [8,9]. Monocular 3D pose methods often assume visible 2D keypoints (e.g. detected by OpenPose or similar) and lift them into 3D. However, in real scenes, self-occlusion or external objects can hide limbs, causing missing or erroneous 2D detections. Fig. 1 illustrates a typical failure mode with the right arm occluded, the corresponding 2D joints vanish, and the right-arm lifting network must be disabled.

Despite the progress achieved by recent occlusion-aware methods, several limitations remain. Many existing approaches depend on static or predefined occlusion patterns, which fail to capture the diversity of real-world occlusions caused by dynamic motion or environmental clutter. Others attempt to hallucinate missing joints or reconstruct them from surrounding cues, but these methods often struggle with severe self-occlusion and unseen occlusion distributions. Moreover, most existing frameworks lack effective adaptation mechanisms during inference, leading to degraded performance when the test-time distribution deviates from the training set.

Furthermore, datasets that assume full or almost complete visibility of the human body are used to train the majority of traditional 3D human posture estimation techniques. Thus, they learn to model the pose space from an innate single-mode distribution that is not subject to fluctuations caused by occlusion. Distributional shift is a phenomenon where the test-time input distribution differs from the training distribution when deployed in real-world scenarios with occlusions. These

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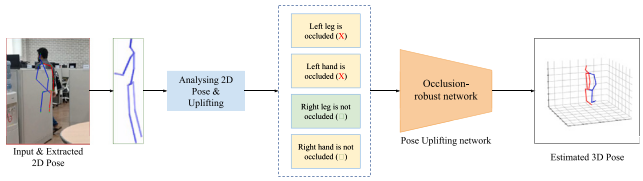

Fig. 1. Conceptual end-to-end process for occlusion robust pose estimation. A partially occluded pose (left hand is occluded) and the corresponding lifting process. The left-side lifting module is marked X since its 2D inputs are missing. As shown, a standard lift-then-complete pipeline cannot operate when keypoints vanish.

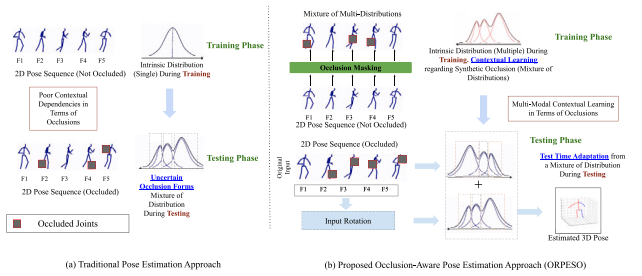

Fig. 2. Differences between existing and the proposed ORPESO. ORPESO captures how poses evolve and learns to aggregate relevant motion information across the sequence, enhancing robustness to individual frame noise or occlusion. For illustrative purposes, a Gaussian distribution is used as a conceptual example. True distribution may differ in the black box transformer network.

models' learned representations do not generalize since they are not trained to manage such variations, which leads to inaccurate joint localization and subpar pose estimation accuracy. For models relying solely on 2D-to-3D lifting, this gap becomes particularly critical, as errors in 3D predictions are directly propagated from missing or corrupted 2D keypoints.

Prior works have explored occlusion-specific strategies by modeling the inherent ambiguity [10], enabling the model to refine noisy, inconsistent poses in occluded frames in video [11], Grap based approach via explicitly forcing the network to become immune to joints being hidden [12] and so on. These approaches indicate that explicitly modeling missing data, adopting multiple hypotheses, or incorporating specialized layers can effectively enhance robustness to occlusion. In this work, these ideas are integrated into a unified 2D-to-3D lifting framework that is intrinsically occlusion-aware.

To address these issues, a novel 3D pose estimation framework, termed Occlusion Robust Pose Estimation via Synthetic Occlusion (ORPESO), is introduced, explicitly incorporating occlusion during training and inference. ORPESO is built on a standard 2D-to-3D lifting backbone. During training, the 2D inputs are augmented by randomly masking (occluding) subsets of joints or image regions, as shown in Fig. 2. During both training and testing, a sample rotation technique combined with prediction averaging is applied for test-time adaptation to further enhance generalization, as illustrated in Fig. 3. ORPESO explicitly models diverse synthetic occlusion patterns during training and introduces a lightweight test-time adaptation mechanism based on rotated-view prediction averaging, enabling the network to recalibrate itself dynamically and maintain robustness under challenging occlusion conditions.

Extensive experiments are conducted on widely used 3DHPE benchmark datasets (Human3.6M and MPI-INF-3DHP) to evaluate the effectiveness of the proposed approach. The proposed approach was shown by the numerical results to outperform the state-of-the-art baselines under a variety of occlusion conditions on Human3.6M, across both Protocol 1 (measured using Mean Per Joint Position Error, MPJPE) and

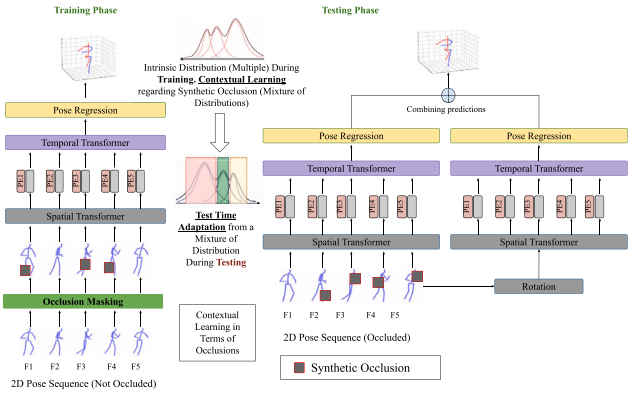

Fig. 3. Overview of the proposed methodology. Occlusion-masking component that teaches the model to selectively ignore occluded regions in input frames. It encourages the network to rely more heavily on unoccluded or complementary temporal information from neighboring frames using a spatio-temporal-based transformer.

Protocol 2 (measured using Procrustes-aligned MPJPE, P-MPJPE). On MPI-INF-3DHP, ORPESO demonstrates consistent numerical improvements over competing techniques, even in difficult real-world situations with dynamic motion and severe occlusions. On the Human3.6M dataset, ORPESO reduces MPJPE by 12.9–17.1 mm (18%–23%) and P-MPJPE by 7.0–11.6 mm (14%–22%) compared to state-of-the-art baselines. On the MPI-INF-3DHP benchmark, it further improves PCK by 5.2–17.3 percentage points, AUC by 2.6–20.6 points, and reduces MPJPE by 12.8–45.0 mm (28%–57%) relative to prior methods. These consistent numerical gains demonstrate the enhanced robustness of ORPESO under occlusion conditions. The main contributions of this work are summarized as follows:

- A novel occlusion-aware 3D pose estimation framework, ORPESO, that augments a standard 2D-to-3D lifting network with synthetic occlusion during training, enabling robust handling of missing joints.
- An implicit mixture-density learning formulation for pose lifting under occlusion. Synthetic occlusion masks are seamlessly incorporated into this training framework.
- A dual-view test-time augmentation and fusion strategy, in which each image is processed in both original and rotated views and the resulting 3D poses are averaged to improve robustness and accuracy under occlusion.
- Thorough experimental evaluation against recent state-of-the-art occlusion-robust methods, demonstrating that ORPESO achieves improved accuracy in both occluded and standard scenarios.

2. Literature review

2.1. Monocular 3D Human Pose Estimation

Recent advances have been made in monocular 3D human pose estimation (3DHPE) using deep-learning-based regression and reconstruction frameworks [13–15]. While more recent methods incorporate temporal consistency and transformer-based designs to better model pose dynamics, early studies like [16] introduced straightforward but efficient 2D-to-3D regression pipelines [17]. To enhance estimation across video sequences, PoseFormer and its offspring, for example, use temporal attention. Likewise, VideoPose3D [18] incorporates temporal convolutions to improve accuracy and smoothness. Developments in the transformer models due to the use of a fisheye camera sensor, such as the work by Zhang et al. [19] and Kundu et al. [20], stretch the limits of temporal modeling by implementing attention processes to

capture long-range relationships. Though these techniques rely on supervised learning over unoccluded training data, the suggested ORPESO approach incorporates a synthetic occlusion-aware training mechanism to explicitly train the model on various occlusion patterns.

2.2. Occlusion-robust 3D pose estimation

The following categories of techniques addressing occlusion in human pose estimation are examined: (a) hallucination-based recovery and (b) occlusion-aware learning [14,15,21]. Despite the increased resilience of these methods, many of them are constrained by static occlusion patterns and lack flexibility during inference. By representing the pose space using a mixture distribution, ORPESO, on the other hand, not only adds a variety of synthetic occlusions to training data but also makes test-time adaptation possible.

2.3. Multi-person and whole-body occlusion-aware methods

Recent efforts have extended occlusion-robust pose estimation to multi-person and whole-body scenarios, which present additional challenges such as inter-person occlusion, dense keypoint distributions, and real-time constraints. For example, POTR-3D [22] proposes a multi-person 3DMPPE approach that improves robustness and temporal smoothness in crowded scenes. STRIDE [23] uses single-video test-time training to refine temporally continuous pose estimates under heavy occlusions. Whole-body and multi-person real-time models (RTMW and related works) further push performance for dense keypoints (face, hands, body) and real-time operation [24,25]. These lines of work suggest promising directions for extending ORPESO to handle inter-person occlusions, and denser keypoint sets.

2.4. Test-time adaptation and mixture modeling

A promising method for improving robustness in domain-shift scenarios, such as occlusion, changing viewpoints, and lighting variations, is test-time adaptation. A number of recent studies have suggested methods for updating the model parameters during inference, including [26,27]. These methods frequently rely on self-supervised loss functions or entropy minimization [28,29]. These methods can be deployed because they are typically unsupervised and do not require ground-truth labels during adaptation. In parallel, mixture models have been used in deep learning to model multimodal outputs and capture uncertainty, as demonstrated by Bayesian deep networks [30] and pose regression using Mixture Density Networks [31]. Through a novel combination of test-time adaptive recalibration and augmented mixture modeling during training, the strategy allows the network to dynamically align with the occluded test-time distribution.

3. Proposed methodology

3.1. Overview

Algorithm 1, and Figs 2, and 3, show the overview of the proposed approach, ORPESO (Occlusion Robust Pose Estimation via Synthetic Occlusion), a unified framework for 3D Human Pose Estimation that can deal with obstructed settings. The proposed method is composed of three fundamental components: (1) occlusion-aware training through synthetic occlusion augmentation, (2) mixture modeling to manage uncertainty and multiple pose hypotheses, and (3) test-time adaptation using self-supervised objectives to dynamically recalibrate to unseen occlusions. The main goal is to enable the pose estimator to generalize well during deployment and model a variety of representations under different occlusion patterns. Using CPN [32] or HRNet [33] to extract a 2D pose sequence from monocular RGB video frames, let $\mathbf{x}_t \in \mathbb{R}^{J\times 2}$ represent the 2D joint coordinates at time step t, where J is the number of body joints. Finding the matching 3D joint coordinates is the goal, where each $\mathbf{y}_t \in \mathbb{R}^{J\times 3}$ represents the 3D positions of the joints. The objective is to learn the mapping function f_θ such that:

$$\hat{\mathbf{y}}_t = f_\theta(\mathbf{x}_t) \quad \text{minimizing} \quad \|\hat{\mathbf{y}}_t - \mathbf{y}_t\|_2^2 \tag{1}$$

Formally, let $\mathbf{x} \in \mathbb{R}^{J\times 2}$ denotes the full 2D joint coordinates, and draw a random binary mask $M \in 0, 1^{J\times J}$ (with each diagonal entry m_j sampled, e.g., from $\text{Bernoulli}(p)$) so that the occluded input is $\tilde{\mathbf{x}} = M\mathbf{x}$. The network takes $\tilde{\mathbf{x}}$ and predicts a mixture of 3D poses to capture ambiguity. Concretely, it outputs mixture weights $\alpha_k(\tilde{\mathbf{x}})$ and mean poses $\boldsymbol{\mu}_k(\tilde{\mathbf{x}})$ for $k = 1, \ldots, K$, modeling the conditional distribution of the 3D pose $\mathbf{y}$ as:

$$p(\mathbf{y} \mid \tilde{\mathbf{x}}) \;=\; \sum_{k=1}^{K} \alpha_k(\tilde{\mathbf{x}})\, \mathcal{N}\big(\mathbf{y}; \boldsymbol{\mu}_k(\tilde{\mathbf{x}}), \Sigma\big), \tag{2}$$

where, Σ denotes a diagonal fixed covariance matrix. Training is performed by maximizing the likelihood of the ground-truth 3D pose $\mathbf{y}^*$ under the proposed mixture model. Similarly, the loss for a batch of N examples is the negative log-likelihood.

$$\mathcal{L}(\theta) \;=\; -\frac{1}{N}\sum_{n=1}^{N} \log \Bigg[\sum_{k=1}^{K} \alpha_k(\tilde{\mathbf{x}}_n) \exp\Big(-\tfrac{1}{2}(\mathbf{y}_n^* - \boldsymbol{\mu}_k(\tilde{\mathbf{x}}_n))^\top \Sigma^{-1} (\mathbf{y}_n^* - \boldsymbol{\mu}_k(\tilde{\mathbf{x}}_n))\Big) \Bigg]. \tag{3}$$

For each training sample, random joint dropping is applied to the 2D input to simulate occlusion, and the model is trained to allocate probability mass across multiple plausible 3D reconstructions. This forces the network to hedge its predictions when faced with missing information, unlike a simple regression, which would confidently predict a single (and often incorrect) pose. When no occlusion is present ($M = I$), the mixture may collapse to one dominant component, recovering standard lifting behavior.

During inference, robustness is further improved through a dual-view augmentation strategy. Given an input RGB image I, a transformed copy I' is also considered, obtained by rotating the camera view by 180° (or another fixed angle). For each view, the corresponding 3D pose is computed as $\hat{\mathbf{y}} = f(I)$ and $\hat{\mathbf{y}}' = f(I')$. The second pose $\hat{\mathbf{y}}'$ is then rotated back to the original coordinate frame by the known camera rotation R (i.e. $\hat{\mathbf{y}}'' = R^{-1}\hat{\mathbf{y}}'$). Finally, the two estimates are combined by averaging:

$$\bar{\mathbf{y}} \;=\; \tfrac{1}{2}\big(\hat{\mathbf{y}} + \hat{\mathbf{y}}''\big) \tag{4}$$

This test-time augmentation (TTA) leverages viewpoint symmetry to reduce sensitivity to occlusion and prediction noise, yielding a more stable final pose estimate. The rotation is implemented as R corresponding to a rotation about the vertical axis (i.e., a 180° yaw).

In summary, the roles of the main components of ORPESO are summarized as follows. Synthetic occlusion augments the training data with diverse and structured missing-joint patterns, enabling the model to learn occlusion-robust pose representations. Mixture modeling captures the inherent ambiguity introduced by occlusions by allowing multiple plausible 3D pose hypotheses rather than a single deterministic prediction. Finally, test-time adaptation improves robustness at inference by recalibrating predictions using rotated inputs and consistency-based fusion, mitigating distribution shifts caused by unseen occlusion patterns. During training, rotation is applied as a data augmentation strategy, while at test time it is used within the dual-view adaptation framework.

3.2. Synthetic occlusion-aware training

By learning pose-relevant cues even when body portions are hidden, this helps the model become resilient to visibility disturbances. With $\mathcal{O}(\mathbf{x}_t, \mathbf{o}_t)$ representing an occlusion operator, the image $\mathbf{x}_t$ is subjected to an occluder $\mathbf{o}_t \sim p(\mathbf{o})$, producing an occluded sample $\tilde{\mathbf{x}}_t = \mathcal{O}(\mathbf{x}_t, \mathbf{o}_t)$. The

Algorithm 1: Training and Inference of ORPESO

Input: 2D joint coordinates x_t, ground-truth 3D pose y_t, rotation operator R, occlusion probability p, trade-off weights α, β, λ.

Output: Trained model parameters θ, estimated 3D pose $\hat{y}_t$.

Training phase:
for *each mini-batch* (x_t, y_t) **do**
  Sample binary occlusion mask $M \sim \text{Bernoulli}(p)$;
  Apply occlusion: $\tilde{x}_t = M x_t$;
  Forward pass through network to obtain mixture parameters $\{\alpha_k(\tilde{x}_t), \mu_k(\tilde{x}_t)\}$;
  Compute negative log-likelihood loss using Eq. (2);
  Combine with occlusion-aware term using Eq. (5): L_{train};
  Update $\theta \leftarrow \theta - \eta \nabla_\theta L_{\text{train}}$;
end
Test-time adaptation:
Generate rotated input $x_t^{rot} = R(x_t)$;
Predict $\hat{y}_t = f_\theta(x_t)$ and $\hat{y}_t^{rot} = f_\theta(x_t^{rot})$;
Rotate prediction back: $\hat{y}_t^{rot_inv} = R^{-1}(\hat{y}_t^{rot})$;
Fuse outputs: $\bar{y}_t = \frac{1}{2}(\hat{y}_t + \hat{y}_t^{rot_inv})$;
Optionally update θ via self-supervised L_{adapt} in Eq. (8);
return $\bar{y}_t$ *as final prediction.*

training loss is defined to take into account both clean and occluded inputs as follows:

$$\mathcal{L}_{\text{train}} = \mathbb{E}_{\mathbf{x}_t, \mathbf{y}_t} \left[\alpha \cdot \| f_\theta(\mathbf{x}_t) - \mathbf{y}_t \|_2^2 + (1-\alpha) \cdot \| f_\theta(\tilde{\mathbf{x}}_t) - \mathbf{y}_t \|_2^2 \right] \tag{5}$$

where the trade-off between learning from occluded and clear views is controlled by $\alpha \in [0, 1]$. This method uses the training data to informally simulate a mixture distribution. The network can generalize across different visibility conditions thanks to the following equation:

$$P_{\text{data}}(\mathbf{x}) = \alpha \cdot P_{\text{clean}}(\mathbf{x}) + (1-\alpha) \cdot P_{\text{occluded}}(\mathbf{x}) \tag{6}$$

3.3. Practical handling of model uncertainty

In practice, a simplified treatment of uncertainty within the mixture model is adopted. The covariance matrix in Eq. (4) is assumed to be *diagonal and fixed*, which provides stable training while avoiding over-parameterization. This assumption effectively models isotropic uncertainty around each predicted mean pose component μ_k. The mixture weights $\alpha_k(\tilde{x})$ are obtained via a softmax layer to ensure valid probability normalization ($\sum_k \alpha_k = 1$). During inference, the final 3D pose is computed as the weighted average of the mixture means:

$$\hat{y} = \sum_k \alpha_k \mu_k, \tag{7}$$

which implicitly integrates model uncertainty across multiple hypotheses. This formulation balances expressive multimodality with computational efficiency.

3.4. Input rotation, and training

Test-time data often contains distributional shifts, such as occlusions or noise patterns not seen during training, which can degrade model performance. To enhance robustness under such conditions, a lightweight test-time adaptation (TTA) strategy from [18] that requires no labels or retraining is incorporated. Given a test input $\mathbf{x}_t$, two versions are generated:

1. The original input $\mathbf{x}_t$
2. A rotated version $\mathbf{x}_t^{\text{rot}}$

The model makes predictions on both versions, and the final prediction is obtained by averaging their outputs:

$$\hat{\mathbf{y}}_t = \frac{1}{2} \left(f_\theta(\mathbf{x}_t) + f_\theta(\mathbf{x}_t^{\text{rot}}) \right), \tag{8}$$

where f_θ is the model with parameters θ. To further refine predictions at test time, self-supervised adaptation is performed by minimizing a loss that aligns the model's output with its own averaged prediction:

$$\mathcal{L}_{\text{adapt}} = \| f_\theta(\mathbf{x}_t) - \hat{\mathbf{y}}_t \|_2^2 + \lambda \cdot \mathcal{L}_{\text{aug}}, \tag{9}$$

where $\mathcal{L}_{\text{aug}}$ is an auxiliary regularization term that encourages consistency across augmented inputs, and λ controls its relative weight. The model parameters are updated using a few gradient descent steps:

$$\theta \leftarrow \theta - \eta \cdot \nabla_\theta \mathcal{L}_{\text{adapt}}, \tag{10}$$

with η as the test-time learning rate. This adaptation procedure allows the model to dynamically adjust to unseen test conditions by leveraging simple augmentations.

3.5. Inference strategy

The overall training objective integrates both the occlusion-aware supervision and augmentation-based regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{train}} + \beta \cdot \mathcal{L}_{\text{aug}}, \tag{11}$$

where β controls the strength of the regularization term $\mathcal{L}_{\text{aug}}$.

3.6. Test-time adaptation

Dual-view test-time adaptation (TTA) is applied during inference. Specifically, both the original and a rotated view of the input are used, and a small number of gradient-based adaptation steps are optionally performed to improve prediction consistency. In this work, S = 3 gradient descent steps per test input are performed using AdamW (learning rate 1e−4, batch size = 1), with updates restricted to the head parameters (BatchNorm affine terms and the final linear layers). To minimize computational overhead, only the head parameters (BatchNorm affine scale/shift and the final linear or mixture layers) are updated, while the backbone remains frozen.

The runtime overhead is quantified using a simple cost model. Let F_{fwd} denote the time for a single forward pass, and let c represent the backward-to-forward cost ratio (empirically $c \approx 2$). The total per-sample latency for dual-view TTA with S adaptation steps can be approximated as

$$T_{\text{total}} \approx [2 + S(1 + rc)]\, F_{\text{fwd}}, \tag{12}$$

where r is the fraction of the backward cost for light updates ($r \approx 0.15$ in implementation).

4. Setup and datasets

4.1. Datasets

Human3.6M [34], contains 3.6 million images of 11 subjects performing 15 daily activities, captured by four high-resolution cameras. It provides both 2D and 3D joint annotations. Subjects S1, S5, S6, S7, and S8 are used for training, while S9 and S11 are reserved for testing.

MPI-INF-3DHP [8] offers greater diversity than Human3.6M, with indoor/outdoor scenes, varied activities, occlusions, and lighting. It includes green-screen and real-world sequences, making it suitable for testing occlusion robustness.

Table 1
Quantitative comparison between SOTA and ORPESO approaches on Human3.6M [34] dataset. Protocol 1, and Protocol 2 indicate the results evaluated using MPJPE, and P-MPJPE metric, respectively. The input 2D poses are detected by CPN [32].

Protocol 1	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Average
PoseFormer [16]	66.8	79.3	73.5	71.0	82.2	73.5	65.9	69.7	69.8	76.1	67.8	77.8	73.4	73.5	101.2	74.8
PoseFormerV2 [17]	63.9	80.3	73.9	61.6	75.4	66.5	62.2	69.8	68.7	71.7	71.1	68.6	71.1	72.8	80.6	70.6
ORPESO (Ours)	**52.9**	**66.5**	**59.5**	**53.9**	**58.5**	**54.3**	**53.5**	**46.5**	**54.6**	**60.8**	**55.1**	**66.2**	**59.2**	**50.6**	**72.8**	**57.7**
Protocol 2	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Average
PoseFormer [16]	48.3	56.8	51.9	52.8	56.9	48.5	47.3	49.9	52.3	55.5	48.2	60.0	55.5	52.6	73.4	54.0
PoseFormerV2 [17]	44.3	54.6	49.7	44.9	52.3	43.4	42.8	48.3	47.5	50.2	46.9	51.9	51.4	52.2	60.4	49.4
ORPESO (Ours)	**38.0**	**47.4**	**42.1**	**41.1**	**43.2**	**38.4**	**38.2**	**32.8**	**40.9**	**43.9**	**39.1**	**51.7**	**44.7**	**37.4**	**56.2**	**42.4**

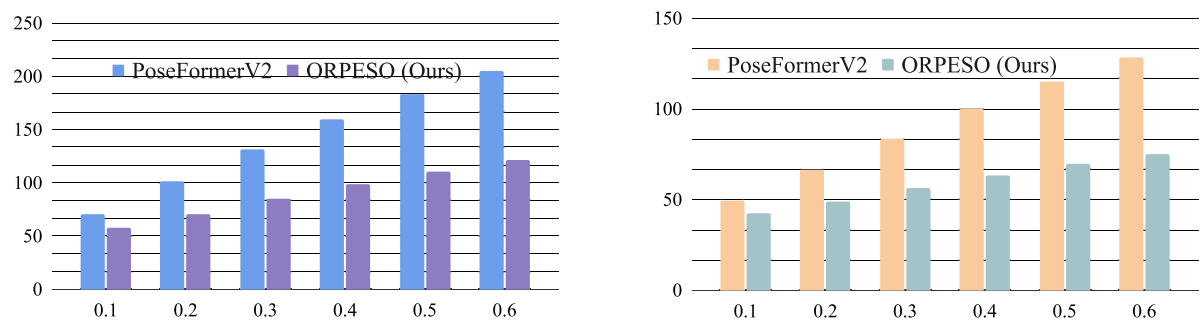

Fig. 4. Performance comparison between PoseFormerV2 and ORPESO (Ours) for different occlusion rates of frames over Human3.6M datasets for Protocol 1 (left) and Protocol 2 (right).

Table 2
Effects of the ORPESO on 2D–3D pose lifting approaches for MPI_INF_3DHP [8] dataset.

Methods	PCK (↑)	AUC (↑)	MPJPE (↓)
PoseFormer [16]	71.4	51.3	78.6
PoseFormerV2 [17]	83.5	69.3	46.4
ORPESO (Ours)	**88.7**	**71.9**	**33.6**

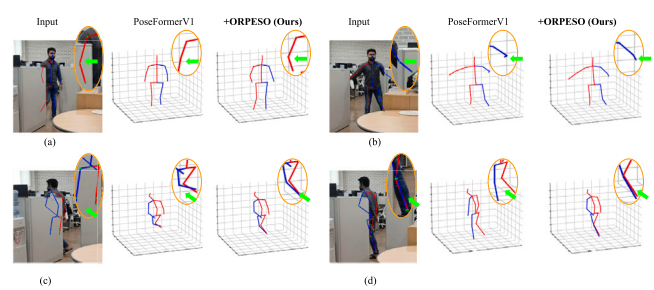

Fig. 5. Qualitative comparison between the baseline and ORPESO versions of PoseFormerV1. Proposed ORPESO improves the performance of the baseline under occlusion for every scenario (a), (b), (c), and (d).

4.2. Experimental setup

The experimental setup and hyperparameters used in PoseFormer [16] and PoseFormerV2 [17] are followed to ensure a fair comparison with ORPESO. To obtain synthetic occlusion, body joints of randomly selected input instances are randomly masked out. All experiments are reported using single-run evaluations following the standard protocol adopted by prior works such as PoseFormer and PoseFormerV2. As the evaluation settings, datasets, and preprocessing strictly follow these baselines, the reported comparisons remain fair and consistent.

4.2.1. Dataset preprocessing and evaluation protocols

The same dataset preprocessing and evaluation setup as PoseFormer [16] are followed, ensuring direct comparability and reproducibility. Specifically, for the Human3.6M dataset, subjects S1, S5, S6, S7, and S8 are used for training, while S9 and S11 are used for testing, adopting the standard 3D pose estimation protocol with MPJPE and P-MPJPE metrics. All images are cropped using the provided bounding boxes, normalized by the camera intrinsic parameters, and temporally aligned to fixed-length frame windows following the PoseFormerV2 settings. To improve robustness against partial visibility, controlled random occlusions are introduced in approximately 10%–60% of the training samples, where a small subset of joints is randomly masked, these occlusion levels are denoted as 0.1 to 0.6 in the figures and analyses. Apart from this occlusion simulation, all other preprocessing steps and evaluation criteria remain identical to PoseFormerV2.

5. Results and discussions

5.1. Comparison with SOTAs on Human3.6M dataset

A quantitative comparison of the suggested ORPESO training strategy versus the most recent state-of-the-art (SOTA) techniques for 3D human pose estimation is presented in Table 1. The popular Human3.6M dataset [34], which covers 15 human actions and reports mean results over test subjects S9 and S11 for further analysis, is used for evaluations. MPJPE and Procrustes-aligned MPJPE (P-MPJPE), which correspond to Protocols 1 and 2, are the two common metrics that are employed. In order to ensure consistency, all approaches are evaluated on a dataset with 10% joint occlusion using 2D poses predicted by the Cascaded Pyramid Network (CPN) [4]. ORPESO achieves lower error than current transformer-based 2D-to-3D lifting models for both protocols and all action categories. In particular, average MPJPE reductions of 17.1 mm and 12.9 mm are observed under Protocol 1 compared to PoseFormer and PoseFormerV2, respectively, along with reductions of 11.6 mm and 7.0 mm under Protocol 2. These findings indicate improved robustness to occlusion compared to prior methods.

Table 3
Computational complexity and real-time performance comparison between PoseFormer and ORPESO on the Human3.6M and MPI-INF-3DHP datasets. All values are per sample. dual denotes original + rotated view, and TTA represents dual-view with test-time adaptation ($S = 3$, light-update).

Model	Mode	Human3.6M		MPI-INF-3DHP	
		GFLOPs	Latency (ms)/FPS	GFLOPs	Latency (ms)/FPS
PoseFormer [16]	Single-view	18.2	30/33.3	16.0	27/37.0
ORPESO (ours)	Dual-view (rotated)	36.4	60/16.6	32.0	54/18.5

5.2. Comparison with SOTAs on MPI_INF_3DHP dataset

The performance results on the MPI-INF-3DHP benchmark, which covers 2D-to-3D pose lifting tasks, are shown in Table 2. Each method is evaluated on the tainted dataset, in which a 10% joint occlusion is applied. To ensure thorough assessment, the evaluation makes use of several standard metrics, including Mean Per Joint Position Error (MPJPE), Area Under the Curve (AUC), and Percentage of Correct Keypoints (PCK). In all tasks, the performance of all tested models is consistently improved by integrating the suggested ORPESO training framework. Regardless of the underlying model architecture or task type, ORPESO produces noticeable gains over baseline training procedures, as evidenced by the findings displayed in Table 2. In particular, ORPESO achieves consistent improvements over a number of SOTA techniques, such as PoseFormer and PoseFormerV2, on the MPI-INF-3DHP dataset. These results highlight the ORPESO approach's general applicability and robustness under occlusion, as reported in Table 2.

5.3. Visual inspection analysis

The qualitative results from visual inspection of projected 3D poses under various occlusion levels are displayed in Fig. 5 to further validate the approach. Even with large joint occlusions, ORPESO produces improved and anatomically consistent poses compared to current techniques. Enhancements are particularly noticeable when self-occlusion or outside obstacles are present, underscoring the model's resilience in difficult, real-world situations.

5.4. Effect of ORPESO for different occlusion rates

The comparative results under varying occlusion rates for Protocol 1 and Protocol 2 are the left and right subfigures in Fig. 4, respectively, show the comparative outcomes for Protocol 1 and Protocol 2 under different occlusion rates. These charts use PoseFormerV2 as a baseline reference to evaluate how resistant the suggested approach, ORPESO, is against increasing visual occlusion levels. For Protocol 1 (left figure), the baseline PoseFormerV2 model exhibits a decline in performance when the occlusion rate rises from 0.1 to 0.5, with MPJPE values increasing from 70.6 mm to 183.5 mm. On the other hand, with values ranging from 57.7 mm to 110.4 mm, the approach continuously maintains a smaller error margin across all occlusion levels.

At increasing occlusion rates, the difference in performance between ORPESO and PoseFormerV2 becomes more noticeable. In particular, a reduction of 73.1 mm relative to PoseFormerV2 is observed at an occlusion level of 0.5. This demonstrates ORPESO's improved robustness to deterioration brought on by blockage. Likewise, ORPESO consistently performs better than PoseFormerV2 for Protocol 2 (right figure) in every tested occlusion condition. While ORPESO maintains lower error levels from 42.4 mm to 69.7 mm, PoseFormerV2's P-MPJPE rises from 49.4 mm at 0.1 occlusion to 115.0 mm at 0.5 occlusion. The performance difference reaches 45.3 mm at the maximum occlusion level, underscoring the resilience of the training approach. In both protocols, the relative performance gains of ORPESO over PoseFormerV2 grow progressively with increasing occlusion.

5.5. Complexity analysis

As summarized in Table 3, ORPESO shares the same transformer backbone as PoseFormerV2, resulting in identical single-view complexity of approximately 18 GFLOPs per sample. Introducing a rotated view doubles the forward computation to 36 GFLOPs, while the addition of test-time adaptation (TTA) with $S = 3$ light gradient steps increases the total cost to roughly 6.2× that of a single forward pass. Empirically, this corresponds to 186 ms latency (about 5.4 FPS) on an RTX 3090 GPU. The dual-view configuration (without TTA) achieves 16 FPS, offering a good balance between robustness and real-time operation for surveillance or motion analysis applications. These results confirm that the proposed rotation and TTA modules improve reliability under viewpoint and occlusion variations with only moderate computational overhead.

6. Conclusion

A robust and adaptable system, ORPESO, is proposed to address the long-standing problem of 3D human pose estimation under occlusions. ORPESO encourages the network to learn a variety of contextual representations by modeling the training data as a mixture of occlusion-induced distributions. ORPESO enhances generalization in real-world scenarios with complicated and structured occlusions through lightweight test-time adaptation during inference without the need for ground-truth annotations. Thorough evaluations on the Human3.6M and MPI-INF-3DHP datasets demonstrate that the proposed approach achieves consistent numerical improvements over state-of-the-art baselines across multiple evaluation metrics and occlusion levels. Future research may explore extending the proposed ORPESO to multi-person pose estimation, integrating domain generalization techniques.

Limitations and Future Works: The test-time adaptation introduces minor latency overhead, which may affect real-time use. This work focuses on single-person 3D pose estimation under synthetic occlusions. In future work, the ORPESO framework will be extended to multi-person and video-based pose estimation for real-time tracking, and indoor and outdoor subsets of MPI-INF-3DHP will be analyzed separately to assess illumination robustness. Multi-person scenarios and crowded scenes, which introduce additional challenges such as inter-person occlusion and identity association, are outside the scope of the current study and are left for future work.

CRediT authorship contribution statement

Md Imtiaz Hossain: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Sharmen Akhter:** Writing – review & editing, Visualization, Software. **Sungjun Yang:** Writing – review & editing, Validation, Resources, Investigation. **Eui-Nam Huh:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.icte.2026.01.013.

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